

Vinh Ly

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Summary

Staff engineer with **18 years of experience** building and operating large-scale distributed systems. Spent the bulk of my career at Google, then moved through Whatnot and YouTube before landing here. I have spent the last several years focused on infrastructure, platform reliability, and the tooling that lets product engineers ship without waiting on ops. Comfortable owning something end-to-end, writing the code, writing the runbook, and handing it off cleanly when the time comes. Languages I reach for most are **Python**, **Go**, and **C++**. Currently looking for a team that treats open-source contribution as part of the job, not a side hobby.

Core Skills

Languages: Python, Go, C++, Java, Bash

Cloud Platforms: AWS (EC2, ECS, Lambda, S3, CloudWatch), GCP (GKE, BigQuery, Pub/Sub), Azure (ACR, AKS)

Infra and Orchestration: Kubernetes, Helm, Docker, Terraform, ArgoCD, Puppet, Ansible

Observability: Prometheus, Grafana, OpenTelemetry, ELK stack, Sentry

CI/CD and Tooling: GitHub Actions, Gerrit, Jenkins, Git

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Own infrastructure for a team responsible for media processing pipelines that handle upload, transcoding, and delivery for YouTube's content creators.
- Migrated a legacy job scheduling system to Kubernetes-based batch workloads. The team stopped needing a dedicated operator for most deploys; on-call pages dropped noticeably as a result.
- Reduced cloud spend by identifying idle compute resources and tightening autoscaling bounds. Most of the savings came from right-sizing ECS task definitions that had been copy-pasted and never revisited.
- Worked with a small cross-functional team of 4 engineers to rebuild the access-control logic for internal tooling. The old system relied on hand-maintained allowlists; the new one pulled from a centralized IAM service, which cut onboarding time for new team members.

Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- Joined as the infrastructure lead for a company in rapid growth. Assessed the existing Kubernetes setup, identified gaps in monitoring, and introduced Prometheus and Grafana as the standard for new services going forward.
- Built a CI/CD pipeline overhaul in GitHub Actions for a monorepo that had grown without a consistent deployment story. Engineers could now deploy any service to staging in under 3 minutes, down from a manual multi-step process that took the better part of an hour.
- Set up GitOps workflows with ArgoCD and established Helm chart conventions across the engineering org. The rollout took about two months and was adopted without mandates because the developer experience was noticeably better than what existed before.

Staff Software Engineer/Manager | Google | Remote | 03/2015 - 04/2024

- Spent the first several years as an individual contributor on large-scale data pipeline infrastructure. Moved into a management role in 2020, overseeing a team of 8 engineers responsible for backend systems serving billions of requests per day.
- Led a redesign of the data ingestion layer for one of Google's internal analytics platforms. The old system had a batch-oriented architecture with end-of-day reprocessing that caused cascading delays. Rebuilt it as a stream-processing system using internal equivalents of Kafka and Flink. Late-night reprocessing runs, which had been routine, became rare.

- Managed the team that owned Google's internal build infrastructure used by thousands of engineers across the company. Reduced average build times from around 12 minutes to under 4 minutes by migrating to distributed caching and parallelizing execution graphs.
- Initiated and drove the adoption of **Python** as the standard scripting language for internal automation tooling. Previously the team used a mix of shell scripts and a custom DSL that nobody outside the team understood. The migration made on-call debugging significantly faster because engineers could actually read the automation code.
- Mentored a cohort of junior and mid-level engineers on systems design and reliability practices. Several were promoted to senior within 18 months; one is now a tech lead elsewhere in the company.
- Collaborated with the security and privacy teams to implement fine-grained audit logging across the pipeline infrastructure. The work was driven by new compliance requirements and took about four months end-to-end.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- Worked on core backend services for Google's internal developer tools. Built and maintained APIs used by thousands of engineers daily for code review, build configuration, and deployment workflows.
- Introduced Puppet for configuration management across a cluster of services that had been managed by hand. The first managed config deployment took longer to write than a manual run would have; the payoff showed up over the next year as the number of services grew and manual config drift became a non-issue.
- Rewrote a C++-based worker service in **Go** because the original codebase had become difficult to maintain and the team needed to ship new features faster. The rewrite took about three months and reduced memory overhead by roughly 40% under load.

Software Design Engineer II | Microsoft | Remote | 10/2007 - 12/2011

- Worked on backend services for a line-of-business product. Owned the data ingestion pipeline that processed roughly 2 million records per day from external sources and loaded them into SQL Server.
- Built an automated testing framework in **Python** that covered the ETL workflows. Previously, validation was mostly manual and error-prone. The framework caught three data-quality issues in its first week of deployment.
- Collaborated with QA engineers and program managers on quarterly release cycles. Responsible for release notes and the internal handoff documentation for the support team.

Software Design Engineer Intern | Microsoft | Hybrid | 06/2006 - 08/2006

- Spent the summer building a prototype internal reporting tool for a product team. The project was small but gave me end-to-end ownership experience from requirements to deployment.

Education

BS Computer Science | University of Texas at Austin | 2003 - 2007